

CryoSand: An Open Trade-Space Sandbox for In-Space Cryogenic Propellant Storage

First Written Report (Progress Report)

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September 25, 2026

Abstract

CryoSand is an open tool for the in-space cryogenic storage trade: when does it pay to refrigerate a propellant tank instead of only insulating it? This report covers the first month of the project. The physics stack (layers L0–L5) has been specified, and build steps 1–7 are implemented, with 54 automated tests passing (analytic limits plus CoolProp-backed closure tests). A first trade has been run with CoolProp properties throughout. Five findings stand out. (i) The optimal cryocooler lift is always either zero or the full heat leak, never in between. (ii) For a vented tank with constant specific masses, the crossover duration has the closed form $t^* = h_{fg} \mu_{\text{act}}$ and does not depend on insulation, tank size or environment. (iii) With the Strobridge cooler-mass correlation, a baseline LEO tank gives $t^* \approx 57$ days for LH₂ and ≈ 1.6 days for LO₂. This is the same order as the published LEO estimates of ~ 64 and ~ 5 days. (iv) The textbook surface-layer closure ($\dot{T}_i = \dot{Q}/m_{\text{surf}}c_p$) violates the ordering the bounding argument needs: at 90% fill it stores *more* energy than the fully mixed tank. It was replaced by an exact two-zone form whose ordering and collapse are now verified for three propellants and four fill levels. (v) Closing the tank shifts the crossover by roughly the hold time. With the exact bounds, the LH₂ crossover spans 57.3 to 377 days depending on the ullage closure, and 1.6 to 921 days for LO₂. So the well-mixed assumption can move the crossover, not just the vent schedule.

1 Problem and objectives

A propellant tank in orbit holds liquid at 20–112 K behind an outer surface at roughly 180–260 K, so heat always leaks in. Each watt that reaches the liquid evaporates it at $\dot{m} = \dot{Q}/h_{fg}$ (for LH₂, 1 W boils off ≈ 0.19 kg/day). A designer can spend mass on insulation and accept the boil-off, or on a cryocooler plus its power system and radiator to remove the heat. The mission duration at which the second option becomes lighter is the *passive/active crossover* t^* . No open tool maps it; the fast storage codes that could be licensed [3].

The project has two objectives: **(E) engineering:** publish a map of t^* across propellant, tank scale, environment and duration; **(R) research:** determine whether the near-universal *well-mixed ullage* assumption moves t^* or only changes the vent schedule. Objective R is answered by a *bounding argument*. Two analytic closures bracket the real pressure rise, and both are carried through the full trade. The spring CFD campaign then refines where real tanks sit between the bounds, but the result does not depend on it.

2 Physical model

The model follows each watt from the Sun to the liquid in six independently testable layers (SI throughout; fluid properties only from CoolProp [2]).

L0 Environment. The outer surface absorbs sunlight, Earth albedo and Earth IR, and radiates to a 4 K sink. Averaged over a sphere's area,

$$q_{\text{abs}} = \alpha \left(\frac{1}{4} G_{\odot} + F_E a G_{\odot} \right) + \varepsilon F_E G_{\text{IR}}, \quad \varepsilon \sigma (T_s^4 - T_{\text{sink}}^4) = q_{\text{abs}}, \quad (1)$$

where $F_E = \frac{1}{2} \left[1 - \sqrt{1 - (R_E/(R_E + h))^2} \right]$ is the sphere-to-Earth view factor. The inward heat leak is less than 1% of q_{abs} , so the outer-surface temperature T_s is set by radiative equilibrium.

L1 Insulation. The multilayer-insulation (MLI) blankets are resistances in series. Struts and plumbing bypass the blankets, so they form a *parallel* conductance:

$$\dot{Q}_{\text{leak}} = \frac{T_s - T_{\text{sat}}}{\sum_i t_i / (k_{\text{eff},i} A) + R_{\text{wall}}} + G_{\text{strut}} (T_s - T_{\text{sat}}), \quad G_{\text{strut}} = \frac{A_s}{L_s} \bar{k}, \quad \bar{k} = \frac{1}{\Delta T} \int k dT. \quad (2)$$

The effective conductivity k_{eff} lumps together radiation, spacer conduction and residual-gas conduction. It is held constant at this fidelity.

L2 Wall. Fourier conduction $\nabla \cdot (k \nabla T) = 0$ is solved by the Summer-2026 finite-difference solver, which uses harmonic-mean face conductivities $k_f = 2k_P k_N / (k_P + k_N)$. At this stage it enters the network as its 1-D limit, $R_{\text{wall}} = t / (kA)$.

L3 Ullage and liquid. Three closures, all first-class:

$$\text{vented: } \dot{m} = \dot{Q}_{\text{net}} / h_{fg}, \quad (3)$$

$$\text{homogeneous (slow bound): } \dot{u} = \dot{Q}_{\text{net}} / m, \quad \rho = m / V, \quad P = P(\rho, u), \quad (4)$$

$$\text{surface (fast bound): } \dot{u}_w = \dot{Q}_{\text{net}} / m_w, \quad \rho_w = m_w / V_w, \quad P = P(\rho_w, u_w). \quad (5)$$

The homogeneous closure is the fully mixed tank, which gives the slowest possible pressure rise. The surface closure puts all incoming energy into a *warm zone*: an interface layer of liquid mass m_{surf} together with the ullage vapour ($m_w = m_{\text{surf}} + m_v$). The rest of the liquid stays at its initial state and occupies a fixed volume, so $V_w = V - (m_l - m_{\text{surf}}) / \rho_l$. While the warm zone is two-phase, its pressure is the saturation pressure of the interface, which gives the fast rise seen in stratified tanks. Because the warm zone is a subset of the tank, $m_{\text{surf}} \rightarrow m_l$ recovers the homogeneous bound exactly. The textbook form $\dot{T}_i = \dot{Q}_{\text{net}} / (m_{\text{surf}} c_p)$ was used first and dropped (§5). m_{surf} / m_l is swept. With $\dot{Q}_{\text{net}}$ constant, each closure is summarised by the energy E_{store} it can absorb before reaching the vent pressure, which gives the hold time $t_{\text{hold}} = E_{\text{store}} / \dot{Q}_{\text{net}}$.

L4 Cooling. The Carnot coefficient of performance is $\text{COP}_C = T_c / (T_h - T_c)$. The specific power is $\phi = W_{\text{in}} / \dot{Q}_{\text{lift}} = 1 / (\eta \text{COP}_C)$, where η is the fraction of Carnot achieved. The radiator rejects $\dot{Q}_{\text{lift}} + W_{\text{in}}$, not $\dot{Q}_{\text{lift}}$ alone: $A_{\text{rad}} = \dot{Q}_{\text{lift}} (1 + \phi) / [\varepsilon \sigma (T_r^4 - T_{\text{sink}}^4)]$. Cooler mass follows the Strobbridge correlation as scaled in [1], $m_{\text{cool}} = 0.2 \dot{Q}^{0.7} [(T_h - T_c) / T_c]^{1.45}$.

L5 Closure. $m_{\text{pen}} = m_{\text{ins}} + m_{\text{cool}} + m_{\text{pow}} + m_{\text{rad}} + m_{\text{lost}}$. The crossover is the duration at which the minimum of m_{pen} over the design space switches from passive to active.

3 Implementation status

The code is in Python and follows three conventions: pure physics functions, frozen dataclasses for state, and a single CoolProp property wrapper. Physics functions receive property values as arguments, so every analytic limit can be tested without a property library. Table 1 lists the build order.

4 Preliminary results

Baseline: a cylinder with hemispherical ends, $r = 2$ m, cylindrical length 3 m ($V = 71.2$ m³, $A = 88.0$ m²), 90% fill at 1 atm. The insulation design unit is a 10-layer MLI sub-blanket. Saturated

Table 1: Build order and status. All listed tests run under `pytest` with CoolProp installed.

Step	Module	Status / verification
1–2	core, network	Done. Series chain matches the hand sum to 10^{-10} ; $k_f(2, 6) = 3$ exactly.
3	L0, L1	Done. View-factor limits $\{\frac{1}{2}, 0\}$; blankets scale as $1/n$; struts verified to be parallel.
4–5	L4, L5, L3 vented	Done. Carnot limits (14, 2.33 W/W); radiator rejects lift plus work; ZBO balance; zero leak gives exactly zero boil-off.
6	model (steady)	Done. First end-to-end run; brute-force optimum reproduces analytic t^* (§4.3).
7	L3 closed bounds	Done. Ordering holds at every m_{surf} for 3 propellants $\times$ 4 fills; collapse exact to 10^{-10} ; energy closes to 0.5%; liquid-full guard.
8	L2 port	1-D form and cylinder $\rightarrow$ plane limit done; 2-D port pending.
9–10	trade , app	First crossover sweep done (Fig. 3); Streamlit interface not started.

properties are computed with CoolProp. All specific masses are the provisional register values of Table 3; the numbers below show the structure of the trade and are not final predictions.

4.1 Environment, heat leak and wall

The absorbed flux (with $\alpha/\varepsilon = 0.32/0.86$) and the resulting surface temperature are 219 W/m^2 and 259 K in LEO at 400 km ($F_E = 0.331$), 109 W/m^2 and 217 K in deep space at 1 AU , and 47 W/m^2 and 176 K in Mars transfer. The wall resistance is $R_{\text{wall}} = 2.3 \times 10^{-6} \text{ K/W}$, or 7.2×10^{-7} of a single 10-layer sub-blanket (3.16 K/W), even with a pessimistic $k = 10 \text{ W/m}\cdot\text{K}$. It is *computed* to be negligible, not assumed so. At the LH_2 ZBO optimum (60 layers, $\dot{Q}_{\text{leak}} = 13.8 \text{ W}$), the struts carry $\approx 9\%$ of the leak with the current placeholder conductance.

4.2 The cooler asymmetry

Figure 1 shows the thermodynamic core of the trade. At the assumed fractions of Carnot, lifting 1 W costs 183 W of electrical power at 20 K but only 11.6 W at 90 K and 8.4 W at 112 K . Every 100 W lifted from LH_2 needs 44.5 m^2 of 300 K radiator, compared with 3.1 m^2 for LO_2 .

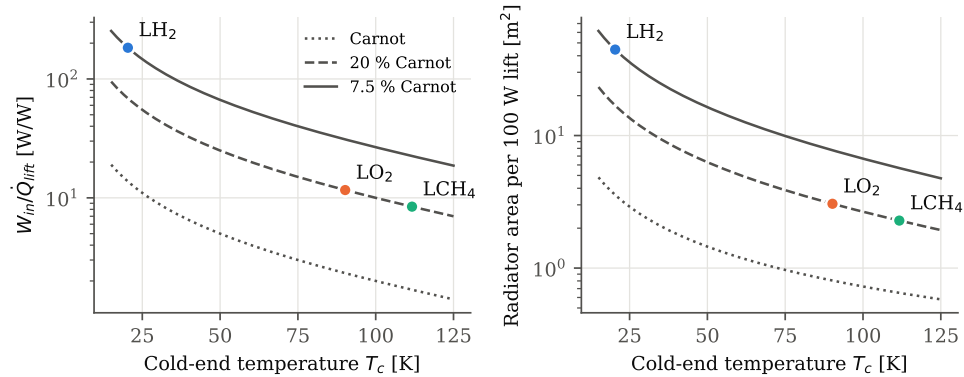


Figure 1: Specific power (left) and radiator area per 100 W of lift (right) against cold-end temperature, for $T_h = T_r = 300 \text{ K}$ and $\varepsilon_r = 0.9$. Markers show each propellant at its assumed fraction of Carnot.

4.3 Structure of the trade and the vented crossover

End-point optimality. For a fixed insulation choice, $m_{\text{pen}}(\dot{Q}_{\text{lift}})$ is the sum of terms that are linear in $\dot{Q}_{\text{lift}}$ (power, radiator, and the boil-off of the uncovered leak) and the concave cooler mass $\propto \dot{Q}_{\text{lift}}^{0.7}$. A concave function on an interval has its minimum at an end point. So the best design is always either fully passive ($\dot{Q}_{\text{lift}} = 0$) or full ZBO ($\dot{Q}_{\text{lift}} = \dot{Q}_{\text{leak}}$). Partial refrigeration is never mass-optimal in this model.

Closed form for constant specific masses. Leaving one watt of leak to boil for a time t costs $\mu_{\text{pas}} = t/h_{fg}$ kg. Removing that watt with a cooler costs $\mu_{\text{act}} = s_{\text{cool}} + s_{\text{pow}}\phi + s_{\text{rad}}(1 + \phi)/q_{\text{rad}}$ kg. Both costs multiply the same leak $\dot{Q}_{\text{leak}}(n)$, so passive wins at *every* insulation thickness exactly when $\mu_{\text{pas}} < \mu_{\text{act}}$:

$$t^* = h_{fg} \mu_{\text{act}}. \quad (6)$$

With constant specific masses, t^* is therefore independent of insulation, tank size and environment; it depends only on the propellant and the refrigeration chain. A brute-force optimisation over n that includes the parallel strut path reproduces Eq. (6) within 2% in the test suite.

With the Strohbridge correlation. Cooler specific mass now falls as $\dot{Q}^{-0.3}$, so larger leaks buy cheaper watts. Figure 2 shows the baseline LEO case: $t^* = 57$ days for LH₂ (ZBO system 255 kg) and 1.6 days for LO₂. LCH₄ gives 2.7 days. Plachta and Kittel [1] found ZBO saving mass beyond ~ 64 days for LH₂ and from as little as ~ 5 days for LO₂ in LEO. The LH₂ agreement is close. The LO₂ result is about 3 times shorter; their MLI model (Lockheed correlation $\times 1.8$) and heat-rejection assumptions differ from ours, and reconciling the two is a register task. Figure 3 is the first crossover map. t^* falls with scale ($70 \rightarrow 48$ days for LH₂, $2.2 \rightarrow 1.4$ days for LO₂, $r = 0.75 \rightarrow 5$ m). The environment changes it by at most 12%, which is what Eq. (6) predicts: scale enters only through the weak $\dot{Q}^{-0.3}$ economy of scale.

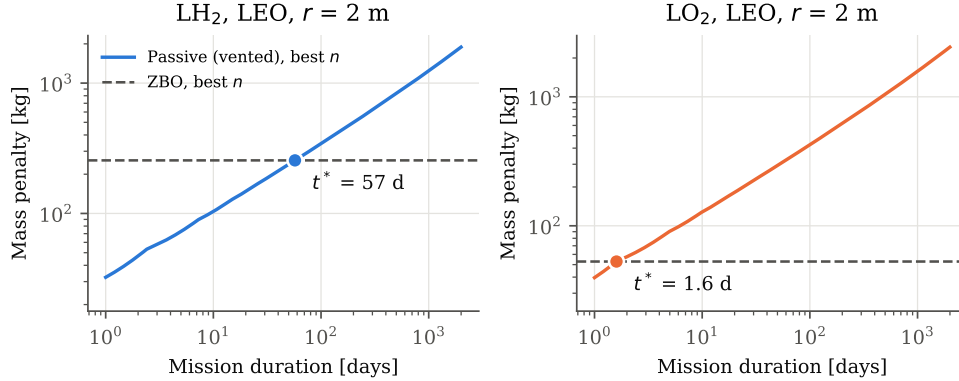


Figure 2: Minimum mass penalty against mission duration for the vented passive design and the ZBO design, each optimised over insulation. The marker is the crossover t^* .

4.4 Closed tank: the ullage closure moves the crossover

A closed tank absorbs E_{store} before its first vent, so for a given insulation choice $m_{\text{lost}} = (\dot{Q}_{\text{leak}}t - E_{\text{store}})^+/h_{fg}$. Repeating the argument above gives

$$t_{\text{closed}}^* = t_{\text{vented}}^* + E_{\text{store}}/\dot{Q}_{\text{leak}} = t_{\text{vented}}^* + t_{\text{hold}}. \quad (7)$$

Eq. (7) turns objective R into a direct comparison: *the ullage assumption matters exactly when the spread in hold time between the two bounds is comparable to t^** . Figure 4 and Table 2 evaluate this for venting at 3 bar, with both closures computed exactly from CoolProp. The homogeneous bound

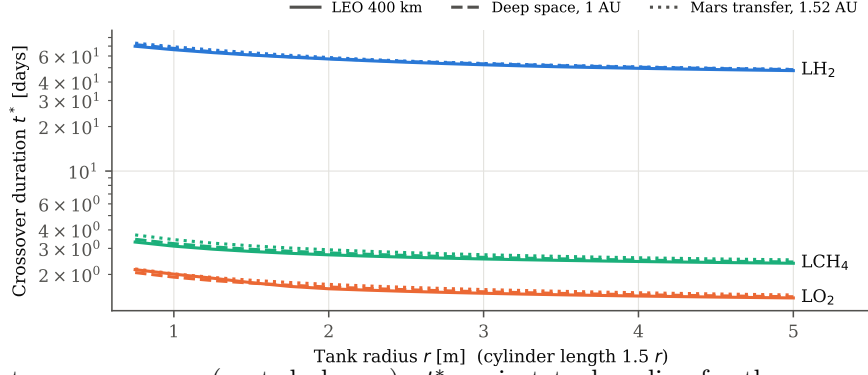


Figure 3: First crossover map (vented closure): t^* against tank radius for three propellants and three environments. The line style marks the environment.

gives hold times of 204 days (LH_2) and 317 days (LO_2). Both are longer than the vented t^* . A thin layer ($m_{\text{surf}}/m_l = 10^{-3}$) holds for only 3.18 and 2.43 days. A conduction-only interface layer (§5, item 1) gives 25 and 24 days. The resulting crossover spans 57.3–377 days for LH_2 and 1.6–921 days for LO_2 : a factor of about 7 for LH_2 and several hundred for LO_2 . Even the physically anchored conduction layer moves the LH_2 crossover from 57.3 to 84.6 days. *First answer to objective R: the well-mixed assumption moves the crossover, not only the vent schedule.* Where real tanks sit inside the band is what the spring work refines. Below $m_{\text{surf}}/m_l \approx 4 \times 10^{-3}$ (LH_2) and 7×10^{-4} (LO_2) the layer evaporates completely before 3 bar (dotted in Fig. 4); condensation back onto the cold liquid is neglected there, which only speeds up the pressure rise, so the curve remains a valid fast bound.

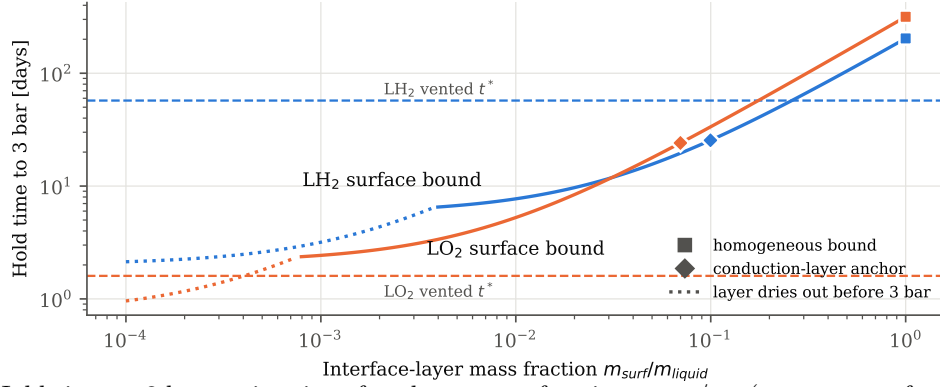


Figure 4: Hold time to 3 bar against interface-layer mass fraction m_{surf}/m_l (two-zone surface bound), with the homogeneous bound (squares, reached exactly at $m_{\text{surf}}/m_l = 1$), the conduction-layer anchor (diamonds) and the vented t^* (dashed). Dotted: the layer dries out before 3 bar. CoolProp properties.

Table 2: Crossover t^* [days] under each closure (LEO, $r = 2$ m, 90% fill, vent at 3 bar). CoolProp properties.

	Vented	Surface, $m_{\text{surf}}/m_l = 10^{-3}$	Surface, conduction anchor	Homogeneous
LH_2	57.3	60.4	84.6 ($m_{\text{surf}}/m_l = 0.100$)	377
LO_2	1.6	7.08	70.1 ($m_{\text{surf}}/m_l = 0.070$)	921

5 Issues identified

1. **m_{surf} needs a physical anchor.** As $m_{\text{surf}} \rightarrow 0$ the surface-bound hold time falls to that of the ullage vapour alone (1–2 days here, Fig. 4), so a sweep alone cannot rule out any answer. A reference scale comes from constant-flux conduction into a semi-infinite liquid [6]: $\Delta T_s = (2q/k)\sqrt{\alpha t/\pi}$, which gives $m_{\text{eff}} = \frac{1}{2}\rho A\sqrt{\pi\alpha t}$. The sweep will be reported around this anchor.
2. **The textbook surface closure broke the ordering (resolved).** `test_bounds_ordered` was written before the L3 physics, and it failed on the first CoolProp run. The c_p -form surface layer at $m_{\text{surf}} = m$ stored 0.4% (LO₂, LCH₄) to 4.4% (LH₂) *more* energy than the homogeneous bound at 90% fill, and 30% less at 30% fill, so it neither ordered nor collapsed. The cause: it follows liquid c_p along the saturation line and ignores the vapour. In a nearly full tank the swelling liquid compresses and condenses the ullage, which releases latent heat, so the constant-volume path needs less outside energy. The two-zone form (§2, L3) applies the exact (ρ, u) closure to the warm zone, so collapse holds by construction and ordering is verified for every m_{surf}/m_l from 10^{-4} to 1. The old form is kept in the code as a regression test that records the failure.
3. **Fill fraction limits the pressure band.** At 90% fill LH₂ is already 98% liquid by volume when it reaches 3 bar. At 95% fill all three propellants become liquid-full first, and the vent pressure is then reached hydraulically. The model now raises an error in that case rather than returning a subcooled state; admissible fill will be a trade variable.
4. **Constant k_{eff} and placeholder specific masses** (power, radiator, struts) dominate the uncertainty in the absolute t^* ; all are flagged in Table 3.

6 Parameter register (Stage 1, in progress)

Table 3: Parameter register. P = provisional (must be sourced before publication); S = sourced.

Quantity	Value	Source / status
$T_{\text{sat}}, \rho_l, h_{fg}$ (LH ₂ /LO ₂ /LCH ₄ , 1 atm)	20.3/90.2/111.7 K; 70.8/1141/422 kg/m ³ ; 446/213/511 kJ/kg	S (CoolProp [2])
Cooler mass	$0.2 \dot{Q}^{0.7} [(T_h - T_c)/T_c]^{1.45}$ kg	S [1]
Fraction of Carnot η	0.075 (LH ₂); 0.20 (LO ₂ , LCH ₄)	P
MLI areal density	0.02 kg/m ² per layer	S [1]
MLI k_{eff} , thickness	3×10^{-5} W/m·K; 25 mm per 30 layers	P
Outer cover α/ε	0.32/0.86	P (confirm in [5])
Power system; radiator	0.025 kg/W; 5 kg/m ² , $\varepsilon = 0.9$, 300 K	P
Strut conductance	5×10^{-3} W/K	P

7 Plan to the December milestone

1. **Done (September):** CoolProp test suite run; bounds reformulated, ordering and collapse verified; all report numbers regenerated from CoolProp.
2. **Oct, week 1:** Sweep the closed-tank crossover across radius, environment and fill (Fig. 3 under all three closures); unify the cooler-mass model between the steady driver and the trade.
3. **October:** Complete the register (power and radiator specific mass, strut $\int k dT$ for G-10 and Ti, MLI $k_{\text{eff}}(T)$), and reconcile the LO₂ crossover with [1].
4. **October–November:** Port the 2-D solver to L2 (axisymmetric geometry, radiative boundary condition via $h_r = \varepsilon\sigma(T^2 + T_\infty^2)(T + T_\infty)$); add fill fraction and vent pressure to the trade.

5. **November:** Validate against published demonstration-tank boil-off (MHTB [4]) to 10%; check energy closure below 0.5%.
 6. **December:** Crossover maps under all three closures; Streamlit interface; public release.
- Risk:** t^* depends strongly on cooler specific mass, so every map will be published with its sensitivity to s_{cool} , η and s_{rad} .

References

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